

Avaneesh Karthikeyan Iyer

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AI Engineer with hands-on experience building real-time voice AI systems, agentic LLM workflows, and audio intelligence pipelines. I'm driven by a passion for solving complex problems through data science and turning advanced technologies into practical, high-impact solutions. With a focus on delivering actionable insights, I thrive in collaborative environments where I can contribute to data-driven innovation and help bridge the gap between research and real-world application.

Work Experience

Agentic AI Specialist : Idolize Business Solutions

Nov 2025 - May 2026

- Built a real-time voice agent system for automated loan follow-ups using LiveKit Cloud, Sarvam AI (Indic language support), and intent detection — reducing manual follow-up effort significantly
- Developed an audio-to-insight pipeline converting PD call recordings into structured credit reports using speaker diarization, Whisper transcription, and LLM-based summarization; output included automated CAS and PD summary documents
- Designed multi-agent LLM workflows in Python to automate loan lifecycle tasks, improving turnaround time on credit operations
- Built and deployed RPA workflows in Microsoft Power Automate for NBFC clients covering contract cancellation and vendor onboarding; monitored and debugged bots to maintain uptime and reliability

HyperAutomation Analyst (Trainee) : Idolize Business Solutions

Oct 2025 - Nov 2025

- Designed and implemented automation workflows in Microsoft Power Automate, with hands-on experience in debugging, workflow optimization, and process improvement.

Internship

AI Engineer Intern: ILM UX Ltd

June 2025 - Sep 2025

- Fine-tuned OCR models for medical transcription using BERT, achieving a 25% accuracy improvement. Additionally, created custom datasets and implemented LSTM-based text classification systems to enhance performance.
- Integrated LiveKit with number providers Twilio, and Airtel to enable seamless real-time voice and video calls using WebRTC and SIP Trunking, successfully testing 25+ calls across platforms. Incorporated Tavus AI for human-like interaction, reducing meeting and onboarding time by 40%.

ML Intern: Herring Infotech

Feb 2025 - May 2025

- Implemented a Load Management System using Reinforcement Learning, Linear Programming, and optimization techniques, incorporating weight and location biases of containers and cartons to enhance space utilization and delivery route efficiency by 45-60%.
- Locally deployed the solution with an interactive Streamlit interface featuring 3D visualizations, delivering scalable and practical logistics management for real-world challenges.

Projects

- **AI-Powered Chatbot:** Designed a real-time, voice-enabled AI chatbot integrating OpenAI's GPT API with speech_recognition and pyttsx3 for seamless speech-to-text and text-to-speech interaction. Implemented context-aware conversational memory to maintain dialogue continuity across multi-turn interactions. Explored latency optimization strategies for local TTS rendering to achieve near-real-time response delivery.
- **Call Intelligence Analyzer:** Built an audio intelligence system to analyze call recordings, generating structured summaries, speaker distribution metrics, and interaction insights using speech-to-text pipelines and LLM-based analysis; implemented MP3/audio conversion and speaker diarization, and experimented with local LLM deployment for improved efficiency and data privacy.

Skills

- **Languages:** Python, SQL (PostgreSQL, Microsoft SQL Server)
- **AI & ML:** Scikit-learn, OpenAI API, LangChain, Hugging Face, Gemini, Azure Document Intelligence, Azure AI (Content Understanding), Prompt Engineering
- **NLP & Generative AI:** LLM-based applications, prompt design, text preprocessing, tokenization, chunking
- **Speech & Voice AI:** Deepgram, Whisper, pyttsx3, SpeechRecognition, speaker diarization, LiveKit, Twilio, Exotel, Tavus
- **Data Science & Visualization:** Pandas, NumPy, Excel, Streamlit, Power BI, Tableau, Plotly, Matplotlib, Seaborn

Educational Background

B.E. Computer Science : SIES Graduate School of Technology, Nerul
CGPA: 8.97

2021-2025

HSC : DAV International School, Kharghar
Grade: 91%

2020-2021

SSC : DAV International School, Kharghar
Grade: 92%

2018-2019

Professional Development & Certifications

- **Google Cloud Skills Boost (Arcade Program)** - 40+ badges in cloud, AI, and machine learning
- **Introduction to Statistics - Stanford University (Coursera)**
- **IBM Data Science - Coursera**
- **Leadership & Volunteering:** Chairperson, CSI SIESGST - Led a 100+ member chapter, organizing national technical events such as project presentations and hackathons, along with workshops and outreach programs to promote knowledge sharing and skill development.